Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 28)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Which of the following oil spill treatments is most likely to kill organisms living deep in beach sediments that were not affected by the oil? Tick one box.

[1 mark]

A Absorbent mops ☐

B Steam washing ☐

C Inflatable booms ☐

D Polymerising agents ☐

0 1 . 2 Explain how polymerising agents make an oil spill easier to clean up.

[2 marks]

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0 1 . 3 Explain why deoxygenation is a more common problem in water than in air.

[2 marks]

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0 2 . 1 Complete Table 1 to explain how each method reduces soil erosion.

[5 marks]

Table 1

Method	How it reduces erosion	
Terracing		
Tied ridging		
Windbreaks / hedgerows		
Mulching		
Strip cropping		

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0 2 . 2 Explain why long-term crops such as tea, coffee or orchards reduce erosion compared with annual arable crops.

[2 marks]

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0 2 . 3 Describe **three** ways livestock can be managed to reduce soil erosion.

[3 marks]

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2 .....

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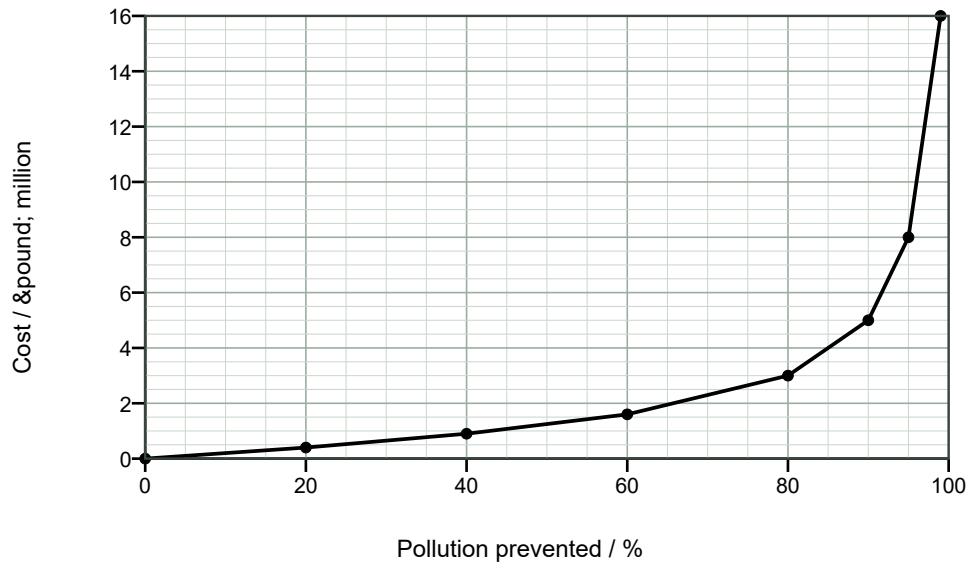
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0 3 Figure 1 shows the relationship between the cost of pollution control equipment at a factory and the percentage of a pollutant emission that is prevented.

Figure 1



0 3 . 1 Describe the relationship shown in Figure 1.

[2 marks]

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0 3 . 2 Use Figure 1 to calculate the extra cost of increasing the pollution prevented from 90% to 99%.

[1 mark]

Answer ☐ £ million

0 3 . 3 Explain what is meant by the terms **ALARA** and **BATNEEC** and how they would be applied by the factory.

[4 marks]

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0 3 . 4 Suggest why the control methods used originally by a factory may no longer be sufficient after several years.

[3 marks]

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0 4 . 1 Give **three** sources of unburnt hydrocarbons in urban air.

[3 marks]

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0 4 . 2 Describe **two** methods used to control hydrocarbon emissions at petrol filling stations.

[2 marks]

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0 4 . 3 Explain how a catalytic converter controls the emissions of carbon monoxide, hydrocarbons and NOx.

[3 marks]

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0 4 . 4 Explain why carbon monoxide is toxic to humans.

[2 marks]

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0 5 Table 1 shows features of three rechargeable battery types.

Table 1

Battery type	Energy density / Wh kg^{-1}	Charge-discharge cycles	Storage efficiency / %	Cost per kWh / £	Safety issues
Lead-acid	35	500	80	100	Toxic lead, acid
Nickel-cadmium	50	1500	75	250	Toxic cadmium
Lithium-ion	200	2000	92	150	Fire risk

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0 5 . 1 Use Table 1 to explain why lithium-ion batteries are used in electric vehicles but lead-acid batteries are still used for stationary storage in some off-grid solar systems.

[4 marks]

0 5 . 2 Give **two** other factors that affect the viability of a rechargeable battery.

[2 marks]

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0 5 . 3 Describe how Vehicle-to-Grid (V2G) systems could reduce the need for peaking power stations.

[2 marks]

0 5 . 4 Explain how Power-to-Gas (P2G) systems use existing infrastructure.

[2 marks]

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0 6 A group of students wants to investigate the effect of vegetation cover on rain-splash soil erosion.

0 6 . 1 Describe how they could carry out the investigation in the laboratory.

[5 marks]

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0 6 . 2 Give **three** variables, other than those you have standardised, that affect rain-splash erosion in the field.

[3 marks]

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0 6 . 3 Explain why it is easier to obtain valid results in the laboratory than in the field, and give **one** disadvantage of laboratory studies.

[2 marks]

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0 7 . 1 Describe the two processes that drive the North Atlantic Conveyor.

[2 marks]

0 7 . 2 Explain how global climate change could weaken the North Atlantic Conveyor and describe the possible consequence for the climate of the UK.

[4 marks]

0 7 . 3 Explain why the effects of climate change on UK temperatures are difficult to predict.

[3 marks]

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0 7 . 4 Describe how deep ocean currents are monitored.

[1 mark]

10

0 8 . 1 Explain how each of the following features of vehicle design reduces energy use.

[6 marks]

Aerodynamics

Reduced mass

Regenerative braking (KERS)

0 8 . 2 Explain why solid steel wheels on rails lose less energy than pneumatic tyres on roads.

[2 marks]

0 8 . 3 Describe **two** ways combustion efficiency in an internal combustion engine can be improved.

[2 marks]

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0 9 . 1 Which of the following is a **secondary** pollutant? Tick one box.

[1 mark]

A Sulfur dioxide from a coal-fired power station ☐

B Tropospheric ozone ☐

C Carbon monoxide from a car exhaust ☐

D Nitrogen monoxide from a diesel engine ☐

0 9 . 2 Explain the difference between a primary pollutant and a secondary pollutant, using acid rain as an example.

[4 marks]

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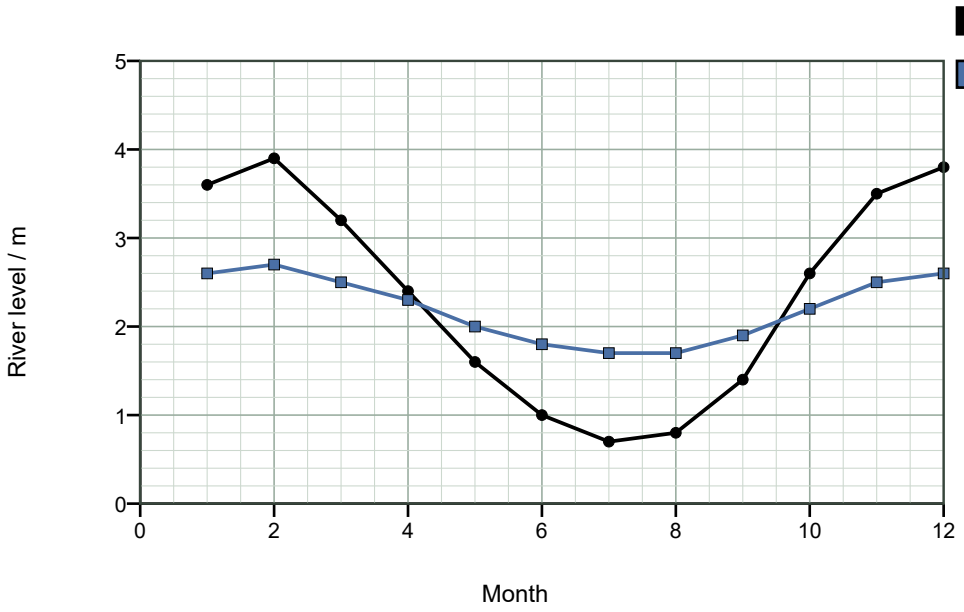
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1 0 Figure 1 shows the river level below a reservoir site during one year, before and after a river-regulation reservoir was built.

Figure 1



1 0 . 1 Use Figure 1 to calculate the reduction in the annual range of river level produced by the reservoir.

[2 marks]

Answer ..... m

1 0 . 2 Explain how the reservoir produces the changes shown in Figure 1.

[2 marks]

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1 0 . 3 Explain why human activities in the catchment may have increased the extremes in river flow before the reservoir was built.

[2 marks]

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1 0 . 4 Discuss the factors that should be considered when selecting a site for a new reservoir, and evaluate the alternative strategies for increasing the sustainability of water supplies.

[9 marks]

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Extra space

15

Write an essay on one of the following topics.

1 1 . 1 Mineral resources are non-renewable. Discuss how an understanding of geological processes and the development of new technologies can extend the period for which society can rely on mineral resources.

[25 marks]

OR

1 1 . 2 Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS